

HARDWARE

- 3x Dell Tower Plus Desktop (OFFICE PCs)
 - \$949.99 per unit, \$2849.97 TOTAL
- 1x Volcora POS terminal
 - \$529.00 per unit
- 6x Dell OptiPlex 5000 Desktop Computer
 - \$899.00 per unit, \$5,394 total
- 1x Cisco MX68CW-HW Meraki MX68CW LTE 802.11ac Router/Security Appliance New
 - \$410.00 per unit
- 1x Ubiquiti UniFi Pro 24 PoE
 - \$699.00 per unit
- 2x Cisco CBW140AC-B 140AC Wi-Fi 5 Wireless Access Point
 - \$115.00 per unit, \$230.00 TOTAL
- 1x PowerEdge T360 Tower Server, 2TB Hard Drive, DDR5
 - \$1252.85
- 1x HP LaserJet Pro MFP 4101fdw Wireless Laser All-in-One Monochrome Printer
 - \$489.00
- Accessories (cables, surge protectors, etc.)
 - \$500

HARDWARE TOTAL: ~\$12,238.82

REASONING: The listed hardware selections were made on the basis of cost-efficiency while also keeping in mind scalability. Customers will primarily be using the PCs within the store to download music or gather information through training classes and will thus not require much computing power; however, choosing a higher-powered PC will allow the business to scale for the future. The PCs used within the office will need to be slightly more efficient to keep workflow smooth and consistent. The infrastructure for the network was chosen with similar logic. The router, switch, access points, and server were all chosen due to their efficiency in relation to their cost. All of the aforementioned devices are compatible with the design of my network. The UniFi Pro 24 PoE was chosen to allow Inter-VLAN Routing via a Layer 3 switch.

Network Design – Physical Layout

There will be three separate VLANs for the office PCs, the guest PCs, and the POS terminal. These VLANs will be segmented by the Ubiquiti UniFi 24-Port PoE Switch. This segmentation is necessary as it prevents customer devices from accessing Point of Sale or office networks.

- Management VLAN: 3 office PCs, server, POS system, printer
- Patron VLAN: Training PCs, access points
- POS VLAN: Separated for transactions

Firewall rules will be configured into the Meraki MX68CW router to only allow necessary traffic between VLANs. For example:

- **Allow** Management VLAN  POS VLAN (for POS transactions)
- **Allow** Management VLAN  File/Print Server (shared files/printing)
- **Allow** Patron VLAN  Internet (web access)
- **Block** Patron VLAN  Management VLAN (protect documents/resources)
- **Block** POS VLAN  Patron VLAN (isolate sensitive POS traffic)

Network Design – Logical Overview

The logical network will be configured through a domain controller, a file/print server, a POS server, and a remote access server. The domain controller will operate through Windows Server Active Directory. The file/print server will serve shared documents and print queues. The POS server will be cloud-based via Lightspeed. For management purposes, MeshCentral will be used to maintain network from off-site.

Logical Overview Cont'd: Redundancy Measures

To avoid any downtime extending beyond 24 hours, the following measures will be taken:

- Cloud-Based POS with High Availability
 - Lightspeed is incredibly reliable and fully redundant allowing us to avoid downtime.
 - We will have a temporary manual solution to process cash transactions in the case of extended downtime (cash register).
- Network Redundancy
 - We will incorporate UPS devices into our critical devices to keep them running during power interruptions
 - Usage of a secondary ISP for automatic failover will also reduce downtime.

Security

- The firewall will block all inter-VLAN traffic except for defined services.
- I will utilize Microsoft Defender for Business as it is designed for small to medium sized businesses. This will provide my business with necessary endpoint protection.
- Everything will be backed up weekly to an external HDD along with a weekly cloud backup.
- MeshCentral will be used for monitoring purposes. It is open source and free of cost.

Software

- Lightspeed for POS system (Robust, cloud-based)
 - \$60/month per store
- Office Suite for office tools
 - \$900/year
- Acronis Cyber Protect for back up solution
 - \$70 per device/year

- MeshCentral for remote management
 - Free
- WooCommerce for e-commerce
 - Free
- Microsoft Defender for Business
 - \$216/year
- SiteGround for web hosting
 - \$240/year

SOFTWARE TOTAL COST/YEAR: ~\$2,146

HARDWARE + SOFTWARE TOTAL = ~\$14,384.82

NETWORK DIAGRAM

